

Chapter 11

Design of Industrial Catalysts

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11.1 Introduction

Catalysts have been used in industry for more than a century, and today over 90% of the chemical manufacturing processes in use throughout the world utilize catalysts in one form or another. Catalysts play important roles in the production of bulk chemicals such as sulphuric acid, ammonia, nitric acid and methanol, and catalysis is essential in hydrocarbon processing such as hydrocracking, hydrodesulphurization, and catalytic reforming. The desire to reduce production cost and energy consumption in industry has been a major driving force for the development of new catalysts and catalytic processes, and in the past decades an increased focus on environmental protection has called for new catalysts for removal of SO₂, NO_x and organic compounds from off-gases in the chemical industry, from power plants, refineries and from auto-exhaust.

Most industrial catalysts are heterogeneous catalysts consisting of solid active components dispersed on the internal surface of an inorganic porous support. The active phases may consist of metals or oxides, and the support (also denoted the carrier) is typically composed of small oxidic structures with a surface area ranging from a few to several hundred m²/g. Catalysts for fixed bed reactors are typically produced as shaped pellets of mm to cm size or as monoliths with mm large gas channels. A catalyst may be useful for its *activity* referring to the rate at which it causes the reaction to approach chemical equilibrium, and for its *selectivity* which is a measure of the extent to which it accelerates the reaction to form the desired product when multiple products are possible [1].

This chapter discusses the steps involved in the development and design of a new SO₂ oxidation catalyst VK69, which was introduced to the market in 1996 by Haldor Topsøe. The strategy and many of the methods are generally applicable to heterogeneous fixed bed catalysts, partly to fluid and slurry bed catalysts, and less relevant for homogeneous catalysts as found in organic synthesis and enzymatic reactions.

The production of sulphuric acid by the contact process, introduced in about 1875, was the first process of industrial significance to utilize heterogeneous catalysts. In this process, SO₂ was oxidized on a platinum catalyst to SO₃, which was subsequently absorbed in aqueous sulphuric acid. Later, the platinum catalyst was superseded by a catalyst containing vanadium oxide and alkali-metal sulphates on a silica carrier, which was cheaper and less prone to poisoning. Further development of the vanadium catalysts over the last decades has led to highly optimized modern sulphuric acid catalysts, which are all based on the vanadium-alkali sulphate system.

11.1.1 Production of Sulphuric Acid

Sulphuric acid is the largest volume chemical in the world with an annual production of about 180 mill. t/year which is used primarily for phosphate fertilizers, petroleum alkylation, copper ore leaching and in smaller quantities for a number of other purposes (pulp and paper, other acids, aluminium, titanium dioxide, plastics, synthetic fibres, dyestuffs, sulphonation etc.). The major sulphur sources for sulphuric acid production are sulphur recovered from hydrocarbon processing in the refineries and from desulphurisation of natural gas, SO₂ from metallurgical smelter operations, spent alkylation acid, and to a minor extent mined elemental sulphur and pyrites. A simplified flow sheet of a modern double-absorption plant for sulphuric acid production from sulphur is shown in Fig. 1.

Sulphur is combusted with dry air at about 1100°C to a gas with typically 10-12% SO₂ and 9-11% O₂. The gas is cooled in a steam boiler to 380-440°C and passed to the converter, where SO₂ is oxidised to SO₃ according to the reversible exothermic reaction



over typically 4 adiabatic fixed beds loaded with sulphuric acid catalyst pellets. Due to the unfavourable position of the equilibrium at high temperature, the gas is cooled in steam superheaters or heat exchangers between the beds.

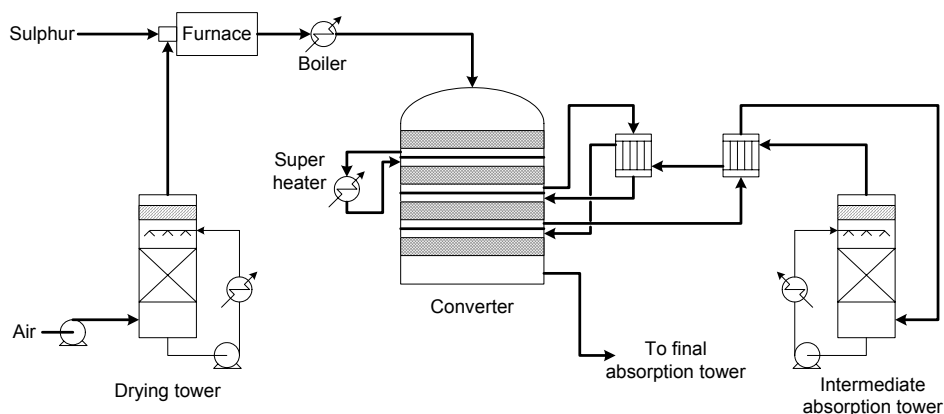


Figure 1. Simplified flow sheet for a sulphur-burning double-absorption sulphuric acid plant (3+1 layout).

Downstream from the 3rd bed, the gas is cooled and passed to an intermediate absorption tower, in which the SO_3 formed is absorbed in recirculating sulphuric acid. The cold and practically SO_3 -free process gas is reheated to 380–440°C and returned to the converter, where the remaining SO_2 is converted to SO_3 in a 4th catalyst bed. The rest of the SO_3 is subsequently recovered in a final absorption tower before the process gas, containing a small fraction of unconverted SO_2 , is emitted through the stack. The combustion air is dried with the 98 wt% product acid in order to avoid corrosion and acid mist problems in the plant. The sulphuric acid process normally operates close to atmospheric pressure with the combustion air blower dimensioned just for compensation of the pressure drop through the plant.

Due to the exothermic nature of reaction (1), the temperature increases through each bed of the converter as SO_2 conversion progresses, cf. Fig. 2. The extent of interbed gas cooling is a compromise between a high approach to equilibrium and a high reaction temperature, both of which will require less catalyst volume. The intermediate absorption enables a total conversion of 99.7–99.9% compared to 97–99% for single absorption plants, dependent on the 4th bed inlet temperature. The layout of plants for processing of smelter off-gases is somewhat different but the converter part is similar to that of the sulphur-burning plant.

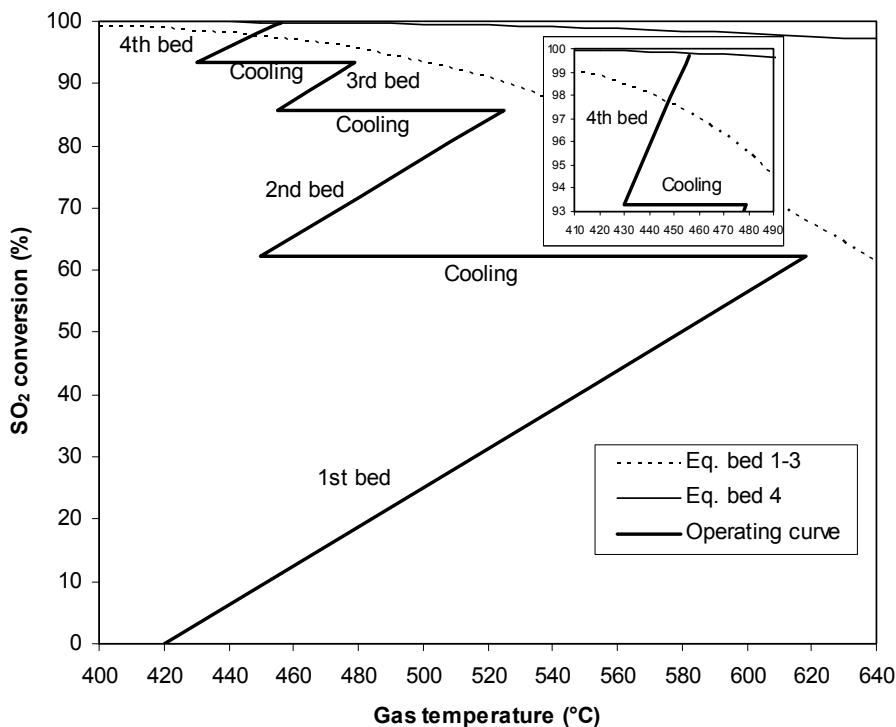


Figure 2. Operating and equilibrium curves showing the total SO_2 conversion for a 4-bed SO_2 converter (3+1 layout) with intermediate absorption of SO_3 downstream bed 3 and conventional catalyst in bed 4. The feed gas contains 11% SO_2 and 10% O_2 .

Commercial sulphuric acid catalysts are based on V_2O_5 dissolved in alkali-metal pyrosulphates on an inactive porous silica support. The carrier is usually made from diatomaceous earth, and the catalyst typically contains 2-5 wt% vanadium and 2-5 mole alkali metal promoter per mole vanadium. The alkali promoter is mainly potassium but sodium and caesium are also used in some catalysts. The catalysts are available as cylindrical pellets, cylindrical rings or finned rings resembling a Daisy or a star as shown in Fig. 3. Generally, in the range of 0.15-0.20 m^3 catalyst is required per ton H_2SO_4 produced per day depending on the actual feed gas, and the service life of the catalyst is typically 5-8 years for 1st and 2nd bed catalyst and above 10 years for 3rd and 4th bed catalyst.

One of the main drivers for the development of new sulphuric acid catalysts over the last decades has been the desire to reduce SO_2 emissions from sulphuric acid plants without costly tail gas cleaning or an additional interbed

absorption step, which would require high investment cost and reduce the energy efficiency of the process. Because of the unfavourable position of the equilibrium at high temperature, this task addresses a serious limitation of the vanadium-potassium-based sulphuric acid catalyst, being a drastic decrease of activity below 400-430°C.

This chapter describes the background, the strategy and the methods used at Haldor Topsøe in the 1990ies to develop and design a new commercial low-temperature sulphuric acid catalyst called VK69.

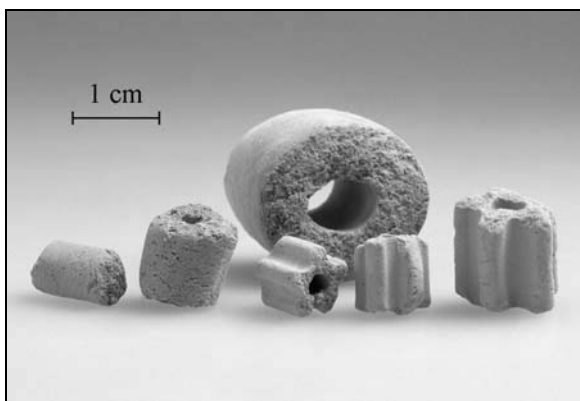


Figure 3. The five variants of size and shape of sulphuric acid catalysts from Haldor Topsøe.

11.2 Design of a New Sulphuric Acid Catalyst

The background for the development of VK69 was a need for reduction of SO₂ emissions from double-absorption plants by installing a more active catalyst at low temperature downstream from the intermediate absorption tower. Clearly, the catalytic solution should be more competitive than the alternatives, e.g. tail gas scrubbing or triple-absorption layout, in terms of capital and operating costs. In the following, the required technical performance of the catalyst with respect to SO₂ oxidation activity, mechanical strength and pressure drop is discussed, and input from the literature and from practical experience in the field is presented. Reviews of the extensive literature published on sulphuric acid catalysts can be found in [2-5].

11.2.1 Catalyst Activity

The activity, which may be defined as the net rate of SO_2 oxidation in moles/s/m³ catalyst according to reaction (1), depends on the intrinsic activity of the catalyst material, the diffusion properties of the catalyst material, the size of the catalyst pellets, and the shape of the pellets.

As a starting point, we will try to quantify the required activity of a new catalyst by simulating the SO_2 emission from a double absorption plant as a function of 4th bed catalyst activity for a feed gas with 11% SO_2 and 10% O_2 and a 3+1 converter with fixed bed volumes, cf. Fig. 4. With a conventional potassium-promoted catalyst such as VK38 from Haldor Topsøe (relative activity 1), a typical requirement in the 1990ies of minimum 99.7% SO_2 conversion corresponding to 395 ppm SO_2 in the stack gas can be achieved in this plant at a 4th bed inlet temperature of 430-435°C. With a 2-3 times more active catalyst in bed 4, the SO_2 emission in this plant can be reduced to below 200 ppm at a lower optimum inlet temperature.

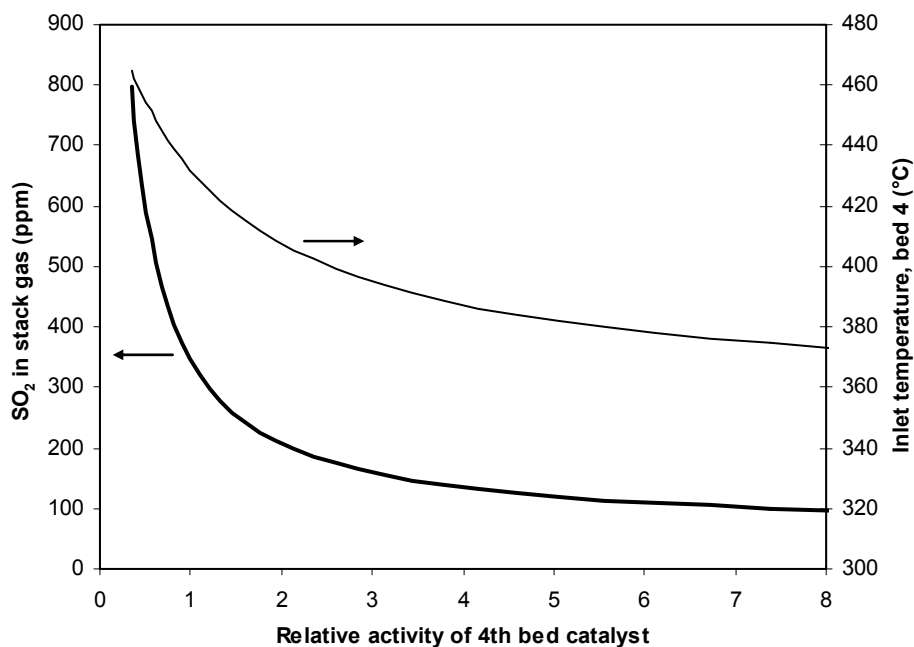


Figure 4. The SO_2 concentration in the stack gas and the optimum 4th bed inlet temperature as a function of 4th bed catalyst activity for a 3+1 double absorption plant with fixed catalyst volume in bed 4.

Sulphuric acid catalysts are not truly heterogeneous catalysts but so-called supported liquid phase (SLP) catalysts, where the oxidation of SO_2 takes place as a homogeneous reaction in a liquid film covering the internal surface of the support material [2]. This was proposed already in 1940 by Frazer and Kirkpatrick [6], who found that the promoting action of the common alkali metals was due to their ability to form relatively low-melting pyrosulphates, which dissolve vanadium oxides, e.g. for potassium



The catalytic activity of the melt was later demonstrated conclusively by Topsøe and Nielsen [7], who obtained a high degree of conversion to SO_3 by bubbling SO_2 and oxygen through a column packed with raschig rings and containing molten potassium pyrosulphate in which 14% wt% V_2O_5 was dissolved. They also prepared catalysts with different alkali metals and concluded that higher activity was obtained if K was exchanged by heavier alkali metals (Cs or Rb) due to their ability to absorb more SO_3 than corresponding to pyrosulphate.

Since then numerous investigations on the co-ordination chemistry of the catalytic melt and the reaction mechanism have been published, but in spite of this the details of the mechanism are still unknown. For many years, the mechanism was assumed to include reduction of vanadium to V^{4+} and reoxidation to V^{5+} by oxygen as proposed by among others Mars and Maessen [8], but in recent years only V^{5+} is believed to be active in the catalytic cycle [5].

However, V^{4+} compounds still play an important role for the activity of the catalyst because an equilibrium exists between V^{5+} and V^{4+} compounds in the melt. The degree of reduction to inactive V^{4+} increases at low temperature and high SO_2 partial pressure, and it has also been found to depend on the liquid dispersion on the support [9]. Furthermore, at temperatures below 500°C some V^{4+} compound precipitates and gradually depletes the melt of V^{5+} when the temperature is lowered, and this partial solidification eventually causes the activity to drop to practically zero at some minimum operating temperature of about 350°C .

Another important parameter is the liquid loading defined as the fraction of the pore volume of the support filled with melt. The activity of a catalyst increases proportional to the liquid loading (which is proportional to the vanadium

content) at low loadings but decreases at high loading due to gas phase pore diffusion restriction and liquid diffusion restriction in the deeper melt pools [10]. As a consequence, an optimum melt loading exists for a given melt composition and operating conditions.

The silica carrier of a sulphuric acid catalyst, which has a relatively low surface area, serves as an inert support for the melt. It must be chemically resistant to the very corrosive pyrosulphate melt and the pore structure of the carrier should be designed for optimum melt distribution and minimum pore diffusion restriction. Diatomaceous earth or synthetic silica may be used as the silica raw material for carrier production. The diatomaceous earth, which is also referred to as diatomite or kieselguhr, is a siliceous, sedimentary rock consisting principally of the fossilised skeletal remains of the diatom, which is a unicellular aquatic plant related to the algae. The supports made from diatomaceous earth, which may be pretreated by calcination or flux-calcination, exhibit bimodal pore size distributions due to the microstructure of the skeletons, cf. Fig. 5.

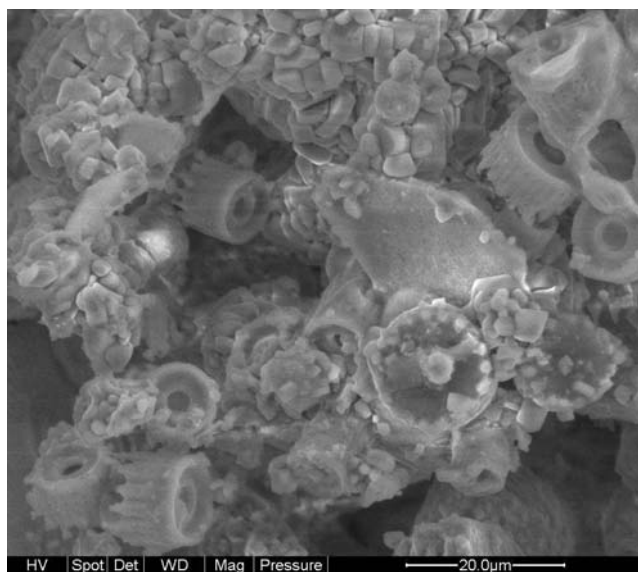


Figure 5. Scanning Electron Micrograph of the interior of a sulphuric acid catalyst. Diatom skeletal fractions more or less covered with solidified catalytic melt are discernible.

The principal idea in the development of VK69 was to use caesium as a low-temperature promoter in a vanadium-based catalyst designed specifically for operation in the beds downstream from the intermediate absorption tower. The company had already gained a lot of experience with Cs in sulphuric acid catalysts because the first Cs-promoted catalyst in the market, VK58, had been introduced back in 1988 by Haldor Topsøe. VK58 was designed primarily for operation in strong gases in the 1st bed and in the final bed of single absorption plants. At that time the available caesium raw material had to be pretreated in the production facility to produce a suitable water-soluble intermediate, but when the VK69 development was initiated, a stable market for water-soluble Cs salts had developed. The complicated chemistry of the vanadium-alkali-metal pyrosulphate catalyst, which is still far from understood, calls for a high degree of empirical development of the chemical composition of the catalyst.

The catalyst activity depends not only on the chemical composition but also on the diffusion properties of the catalyst material and on the size and shape of the catalyst pellets because transport limitations through the gas boundary layer around the pellets and through the porous material reduce the overall reaction rate. The influence of gas film restrictions, which depends on the pellet size and gas velocity, is usually low in sulphuric acid converters. The effective diffusivity in the catalyst depends on the porosity, the pore size distribution, and the tortuosity of the pore system. It may be improved in the design of the carrier by e.g. increasing the porosity or the pore size, but usually such improvements will also lead to a reduction of mechanical strength. The effect of transport restrictions is normally expressed as an effectiveness factor η defined as the ratio between observed reaction rate for a catalyst pellet and the intrinsic reaction rate, i.e. the hypothetical reaction rate if bulk or surface conditions (temperature, pressure, concentrations) prevailed throughout the pellet [11]. For particles with the same intrinsic reaction rate and the same pore system, the surface effectiveness factor only depends on an equivalent particle diameter given by

$$d_{eq} = 6 \cdot \frac{V_p}{A_p} \quad (3)$$

where A_p is the external surface area and V_p the volume of the particle. A calculation of the effectiveness factors for a conventional catalyst without caesium in the 4th bed of a 3+1 converter as a function of d_{eq} shows the benefit of small catalyst particles under this condition (Fig. 6). For a more active catalyst material than that in Fig. 6, the effectiveness factor will be lower for the same d_{eq} . For very active catalyst systems, the active components can be utilized more efficiently by shell-impregnation or as monoliths, but this was not considered advantageous for VK69.

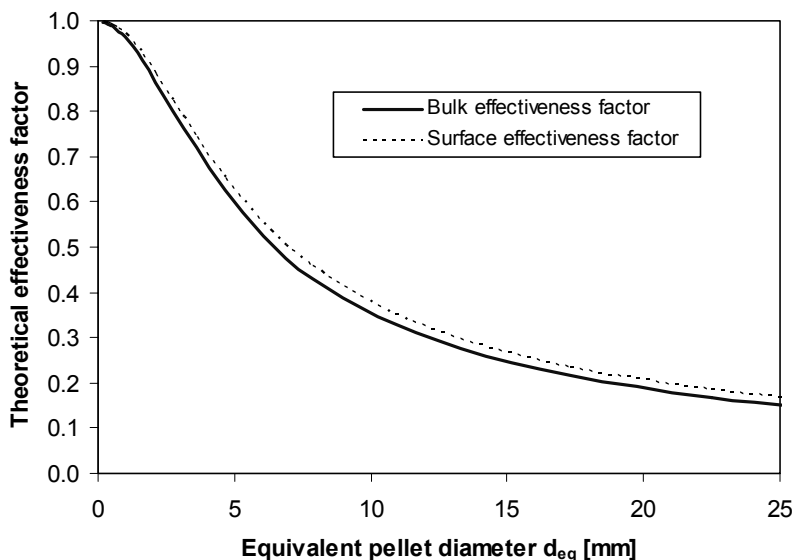


Figure 6. Effectiveness factor for SO_2 oxidation over VK38 in 0.7% SO_2 and 7% O_2 at 400°C.

Another important constraint especially for existing industrial converters is the volume available in the reactor for the catalyst. Accordingly, the reaction rate per unit bed volume

$$r_{\text{SO}_2, \text{bed}} = (1 - \varepsilon) \cdot r_{\text{SO}_2, \text{pellet}} = \eta_b \cdot (1 - \varepsilon) \cdot r_{\text{SO}_2, \text{intrinsic}} \quad (4)$$

must be considered. A very open-structured catalyst, e.g. a thin-walled monolith, with a high bed void fraction ε may have a high effectiveness factor but a much higher reactor volume will be required to provide the same total conversion. Likewise, small pellets providing a high effectiveness factor will lead to a high pressure drop across the catalyst bed which can be unacceptable (cf. section 2.3 below).

11.2.2 Mechanical Strength

The mechanical strength of a catalyst is essential for maintaining low pressure drop over the converter. The first requirement is a low fractional break-up during handling, transportation and loading in the converter, where the pellets

fall from a height of typically 0.5-1.5 m. Furthermore, the pellets must be able to resist the forces imposed on the catalyst in a fixed bed including the weight of catalyst above, the force applied by the pressure drop, and the movement due to thermal expansion of the converter at varying temperatures. Finally, the pellets will experience pneumatic transport because sulphuric acid catalyst beds are normally screened regularly in order to remove dust accumulated in the bed from the synthesis gas. Prediction of screening loss of a new catalyst is complicated by the fact that the pellet strength depends on the history of operating conditions experienced by the catalyst in the converter.

In addition to the external forces, the catalyst must also resist internal forces imposed on the pellet as phase transitions in the catalyst material progress. These transitions, including e.g. transformation of the amorphous silica carrier into crystalline α -cristobalite, precipitation of V^{4+} and V^{3+} compounds, and destruction of the carrier by the melt, may eventually cause the catalyst to break up in smaller particles or even to catalyst powder.

The catalyst strength is generally improved at low porosity but this usually causes an adverse effect on the activity. Due to the low operating temperature and low SO_2 concentration, the conditions for the new catalyst in a 4th bed are not as severe as in the upper beds.

11.2.3 Pressure Drop

The desire to save energy calls for low pressure drop over the catalyst layers because they account for a significant part of the total pressure drop through the sulphuric acid plant. According to simple correlations such as the Ergun equation [12], the pressure drop over a catalyst bed per bed length at a given flow rate and properties of the gas only depends on the bed void fraction ε and a characteristic pellet diameter

$$\Delta P = k \cdot \frac{1}{d_{eq}} \cdot \frac{1 - \varepsilon}{\varepsilon^3} \quad (5)$$

where k is a proportionality constant and d_{eq} is given by (3). For a reaction rate controlled by pore diffusion, the pellet reaction rate is inversely proportional to d_{eq} , and as a consequence of (5) the ratio of catalyst activity to pressure drop is independent of pellet size for fixed geometrical shape in this case. In practice it must be taken into account that the void fraction depends both on industrial loading procedure and on the distribution of pellet lengths from the extrusion process.

11.2.4 Production

In principle, any catalyst developed in the laboratory can be manufactured in large scale. In practice, however, the necessary investments, which may include development of the production process, and the operating costs of the catalyst production plant including availability and cost of raw materials, plant maintenance, labour etc. may not be justified by the market potential. In the VK69 case, production in the existing plant according to the route in Fig. 8 was preferred but not a strict requirement.

11.3 Design Strategy

The development and design of the new sulphuric acid catalyst involves numerous disciplines, work tasks and methods from the field of practical catalysis and chemical engineering. The most important steps of the development process are briefly described in this section but for a deeper technical approach to the theory and methods, the reader is referred to text books within the field [1, 13-14].

The starting point for a route to a new commercial catalyst is typically new market opportunities or requirements to performance identified in a competitive market, cf. Fig. 7. However, ideas fostered from observations in the laboratory or elsewhere in an organisation may also be recognised as a source of new product opportunities. In the first stages of the catalyst development, formulation with different carriers, shapes, compositions and calcinations are tested with respect to catalytic activity, mechanical strength and pressure drop. In the later stages of the development, full-scale production is demonstrated, the catalyst stability and durability are validated and detailed performance data are acquired. An innovative attitude is important especially in the first stages of development, where different catalyst formulations are proposed and selected, but in all steps of the testing much can be learned from the experimental observations. In the evaluation steps shown in Fig. 7, the profitability of the catalysts is compared based on the measured performance, estimated production cost and market analyses. However, the profitability should be taken into account throughout the development process in order to avoid e.g. exotic production routes or expensive raw materials that will fail to pass the evaluations anyway. In this way, the work on non-competitive products can be stopped early in the process. The individual tasks and methods used for the development of VK69 are described in the following and a selection of results are presented and discussed in the next section.

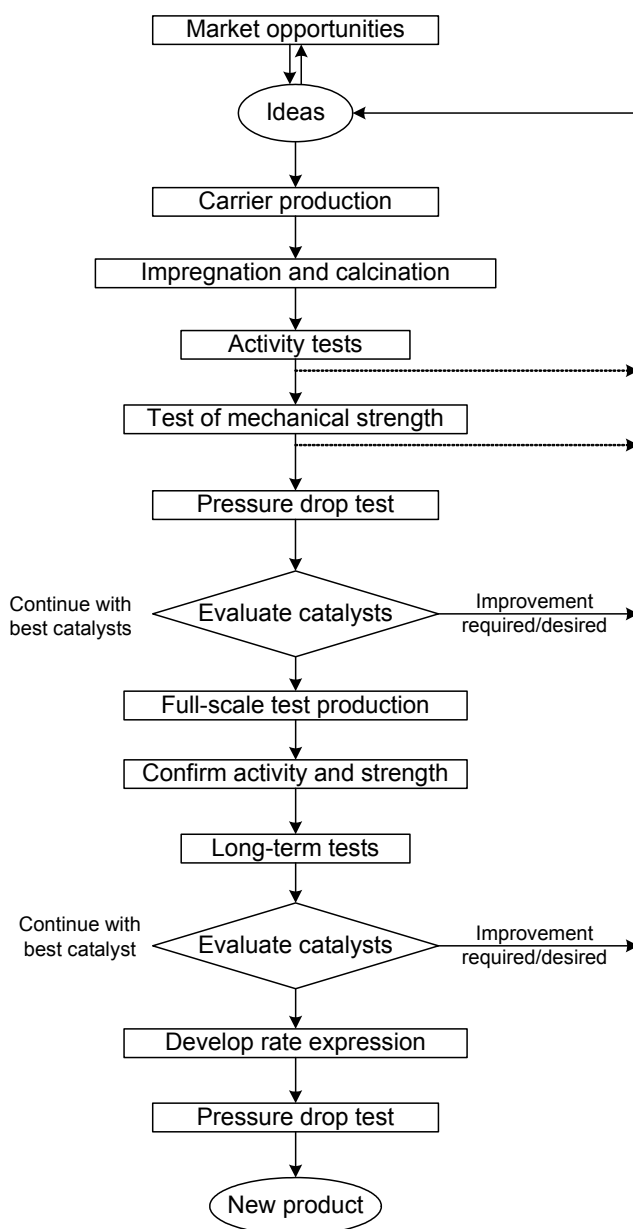


Figure 7. Strategy for design of a new sulphuric acid catalyst.

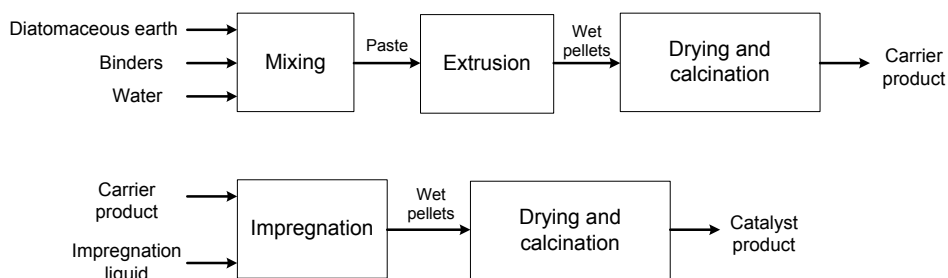


Figure 8. Simplified diagram of unit operation in the production of sulphuric acid catalyst.

11.3.1 Production of Carrier

The first task was to produce carriers from different recipes and in different shapes as shown schematically in Fig. 8. The raw materials diatomaceous earth, water and various binders are mixed to a paste, which is subsequently extruded through a shaped nozzle and cut off to wet pellets. The wet pellets are finally dried and heated in a furnace in an oxidising atmosphere (calcination). The nozzle geometry determines the cross section of the pellet (cf. Fig. 3) and the pellet length is controlled by adjusting the cut-off device. Important parameters in the extrusion process are the dry matter content and the viscosity of the paste. The pore volume distribution of the carriers is measured by Hg porosimetry, in which the penetration of Hg into the pores of the carrier is measured as a function of applied pressure, and the surface area is measured by the BET method, which is based on adsorption of nitrogen on the carrier surface [1].

11.3.2 Impregnation and Calcination

The carriers were impregnated to different compositions in terms of vanadium content, K/V ratio, Na/V ratio, Cs/V ratio and sulphur content, cf. Fig. 8. The exact sulphur content is less important since the melt, according to reaction (2), takes up an equilibrium amount of SO_3 corresponding to a sulphur to alkali metal molar ratio of about 1 during operation in synthesis gas. The impregnation was carried out by submerging the pellets for a period of time in a surplus of aqueous impregnation liquid prepared from sulphuric acid and water-

soluble salts of V and alkali promoters K, Na, and Cs. In this step, the liquid is sucked into the pores of the carrier due to capillary forces and the final catalyst composition is determined by the porosity of the carrier and the concentrations in the impregnation liquid. The wet pellets were finally dried and calcinated by heating in a furnace in an oxidising atmosphere.

The catalyst samples produced as described above were subsequently analyzed and tested in the laboratory with respect to catalytic activity, mechanical strength and pressure drop.

11.3.3 Activity Test

Because the composition and nano-scale structure of a catalyst depend on the temperature and the gas environment, it is important to measure catalyst activity under conditions experienced in the industrial converters. For the sulphuric acid catalysts, the activity for conversion of SO_2 to SO_3 was measured in the set-up shown in Fig. 9.

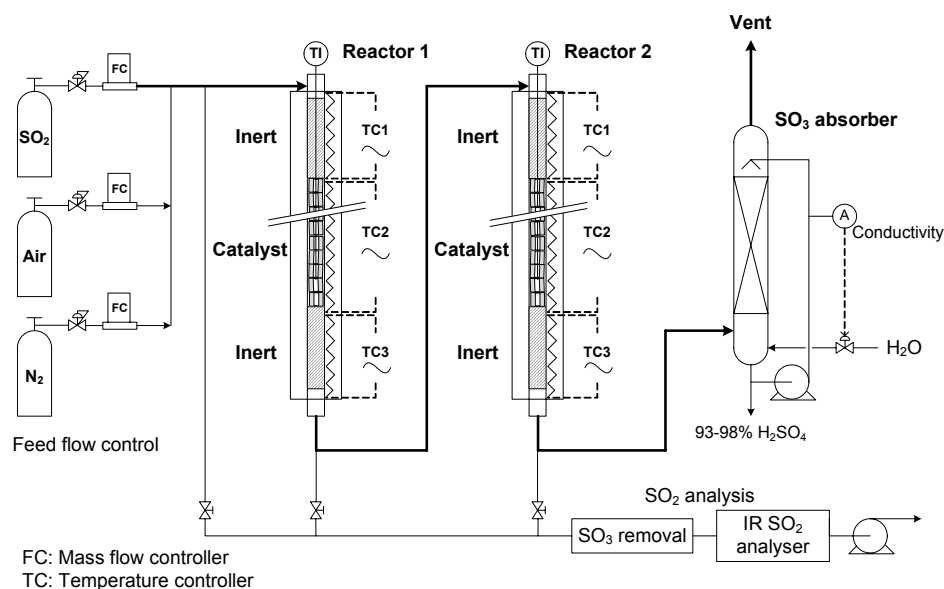


Figure 9. Experimental unit for measurement of catalytic SO_2 oxidation activity.

A dry feed gas is mixed from SO_2 , air and nitrogen to a total flow of 50-200 Nl/h with typically 0.2-20 % SO_2 and 3-20 % O_2 . For wet gases or O_2 -enriched conditions, water vapour and oxygen can also be added to the feed gas. The conversion of SO_2 over the catalyst is measured in one or more isothermal integral fixed bed reactors connected in series and operating typically at 300-750°C with 10-90% incremental SO_2 conversion in each reactor (2 reactors are shown in the figure). Isothermal conditions in the catalyst bed are achieved by controlling the output to the three heater zones of the reactor oven. The effluent gas from the final reactor is passed to an SO_3 absorber, since the SO_3 would otherwise produce sulphuric acid mist in the environment when mixed with humid air. The SO_3 is absorbed in recirculating concentrated sulphuric acid, the concentration of which is automatically controlled by conductivity measurement. The conversions are determined by an IR SO_2 analyser as shown in the figure or by titration with iodine (Reich method). Because isothermal measurements do not follow the adiabatic reaction lines shown in Fig. 2, several measurements at different temperature and conversion levels may be necessary to adequately cover the conditions through an industrial adiabatic bed.

Various test reactors are available for the measurement of catalytic activity (adiabatic integral reactors, isothermal integral reactors, differential recirculation reactors, Carberry reactors, Berty reactors etc.), and the catalyst loading can be designed to measure the activity of whole pellets or intrinsic activity (small particles). The reactors used in the present case are so-called single pellet string reactors (SPS), in which a number of whole pellets are stacked on top of each other in a single row in a reactor tube at isothermal conditions. This reactor type, which has previously been investigated for reaction between pelletized iron ores and gases [15], has been adopted and perfected by Haldor Topsøe for catalyst activity measurements. For sulphuric acid catalysts, we have found that the SPS reactor is an excellent approximation to large scale adiabatic reactors while maintaining the advantages of small-scale units, e.g. the ability to operate isothermally, quick response and ease of operation. The advantage of using whole pellets is that the measured reaction rate includes all effects of transport and inhomogeneities inside the pellet and hence no additional uncertainty is introduced from interpretation of internal transport restrictions. Catalyst samples may directly be ranked according to measured conversions under equal conditions, but further quantification requires some reaction rate model because the integral conversion is measured. For other catalyst systems involving a very active catalyst or a very high heat production in the pellet, external transport restrictions are more important and in this case, more attention must be paid to correcting for differences in mass and heat transfer coefficients between the SPS reactor and industrial reactors.

The initial screening of VK69 catalyst samples was carried out at 380-420°C in 0.7% SO₂ and 7% O₂ which is a typical 4th bed condition in a 3+1 double absorption plant. A lot of the samples, however, were also tested under other conditions partly to test their performance in a broader range, e.g. 3rd bed in a 2+2 layout, and partly to keep an eye open for beneficial effects of the formulation not detected at the reference conditions. The initial testing also provided information about catalyst stability. In general, the measurements are preferably based on stable activity but some catalyst systems deactivate to a stable level during initial operation while others keep deactivating. If deactivation occurs, it is important to compare the catalysts after equal periods on stream. On the other hand, if the catalyst activity continuously decays during the test, stabilisation of the carrier or the active phases may turn out to be the focal point of the product development.

Based on the activity test results, new experimental series with different impregnations on the same carrier are typically carried out in order to optimise the chemical composition for a given carrier.

11.3.4 Test of Mechanical Strength

Several methods have been used for quantification of mechanical strength of catalysts and carriers since no single method simulates all the requirements mentioned previously. The *attrition loss* is measured by placing a quantity of pellets in a rotating horizontal cylinder. After a number of revolutions, the attrition loss is determined by sieving smaller pieces from the sample. The *bulk crush strength* is measured by slowly forcing a piston down onto a bed of catalyst pellets and measuring the displacement as a function of applied pressure. The *side crush strength* is a measure of the radial force necessary for a piston to crush single pellets. In the *drop test* a number of pellets are dropped individually from a height of 12 m and the number of broken pellets and the weight lost by fragmentation are recorded.

Although the results from these methods may correlate to some extent, they simulate different essential requirements of the catalysts. Attrition loss primarily relates to handling, transport, loading and screening whereas the crush tests simulate the forces imposed on the catalyst in a fixed bed. The drop test simulates the risk of catalyst break-up during loading and pneumatic transport.

When these tests are used for both carrier and catalyst, the catalyst is stronger than the carrier due to the vanadium and alkali salts, which may constitute 30-40% of the catalyst weight. However, the increased strength measured for the

pellets after impregnation cannot be expected truly to represent all aspects of the industrial conditions, because the salts are in a molten state during operation and hence contributes less to the pellet strength.

11.3.5 Test of Pressure Drop

The pressure drop for the produced carrier or catalyst is measured in the setup shown in Fig. 10. An adjustable flow rate of 300-700 Nm³/h ambient air is supplied by a blower and passed downwards through a bed of catalyst in a long tube. The diameter of the bed is 0.39 m, which is well above the minimum of 10 pellet diameters required for satisfactory reproduction of the void fraction observed in a large fixed bed. The catalyst is poured into the tube from the top and the bed may subsequently be settled by applying a reproducible tapping or vibration to the tube. Since the latter reduces the void and increases the pressure drop, it is important that the catalysts are loaded and vibrated in the same way in order to get comparable results. The pressure drop without catalyst should be checked in order not to introduce errors from the support grid or measuring taps.

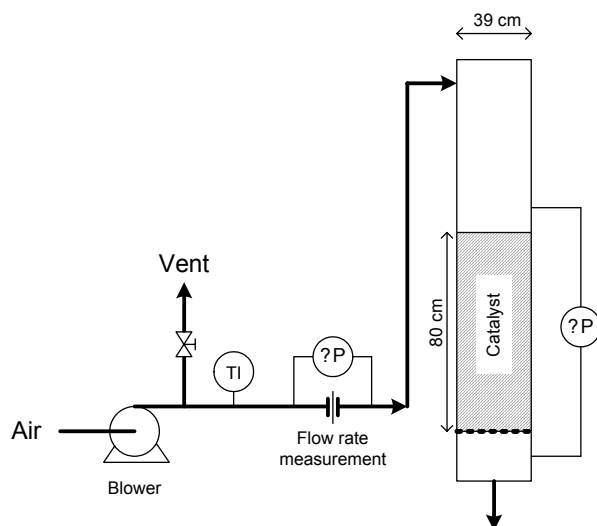


Figure 10. Experimental unit for measurement of pressure drop over a catalyst bed.

The flow pattern is primarily determined by the particle Reynolds number, which is about 100 in the industrial converter (superficial velocity 0.35-0.55 Nm/s), but in order to improve the accuracy of the comparison of different catalysts, higher flow rates are also included. Although theoretical correlations can be used for extrapolating the measured pressure drops for a new shape to the industrial operation temperature, a more reliable method is to calculate the pressure drop from industrial experience for well-known shapes, e.g. 10-mm ring, and assume the same relative pressure drop as in the cold measurements.

11.3.6 Full-scale Test Production

After the evaluation of the performance of the different catalysts prepared in laboratory and pilot scale, a few candidates are selected for a full-scale test production. The main purpose is to demonstrate that the production steps shown in Fig. 8 are feasible for the new catalyst and that a uniform and satisfactory product quality can be achieved at an acceptable production rate in the existing production line.

Normally, the best activities observed during the development process are found for catalysts prepared in the laboratory where special attention is paid to each preparation step and where better control of e.g. impregnation and calcination temperature history can be achieved. This should be kept in mind when comparing activities of new lab-prepared catalysts with standard products from a production facility. As a consequence, it is seldom worth the effort to continue with a test production if the activity of the lab-prepared catalyst fails to meet the requirements. Important results for the test-produced catalysts are activity measurements covering the full range of operating conditions in the industrial converter and the mechanical strength.

In some cases, it may not be possible to produce the new catalyst in an existing production facility due to e.g. limitations of the existing equipment or a need for new unit operations. In this case, a pilot scale production or a full-scale demonstration of the critical steps should be considered before investing in a new production line.

11.3.7 Test of Deactivation

Already in the first activity tests, information on the rate of deactivation is gained because the SO₂ conversion is measured several times during an activity

test running from a couple of days to some weeks until stable activity is observed. However, at some point during the development process, catalyst deactivation and stability must be tested under the relevant operating conditions in a long-term test running for extended periods, e.g. a year. This may be carried out under well-controlled conditions in a dedicated laboratory setup similar to the one in Fig. 9, possibly with reactors connected in parallel with different catalysts. For a more rigorous test, an open canister with the catalyst can be placed in the full depth of an industrial catalyst bed for a year or two and subsequently be analysed in the laboratory. The strength of this test is that the catalyst experiences the industrial gas including impurities such as dust and potential catalyst poisons, and that several start-ups, shut-downs, temperature excursion and concentration variations may be included in the history of the test.

11.3.8 Development of Rate Expression

When the best catalyst has been chosen and found to fulfil the requirements with respect to activity, strength, pressure drop, production and profitability, a procedure must be developed for calculation of the catalyst volume required to obtain a given SO₂ conversion in an industrial reactor. In its simplest form, the calculation basis can be a table or an expression for space velocity (NHSV) as a function of feed gas properties and final conversion. A more detailed approach is used for design of catalytic reactors at Haldor Topsøe, where a rate expression of the form

$$\text{reaction rate} = -r_{\text{SO}_2}(p_{\text{SO}_2}, p_{\text{O}_2}, p_{\text{SO}_3}, P, T) \quad (6)$$

is integrated through the fixed bed by a proprietary reactor simulation tool. The detailed form of r_{SO_2} , which of course must include the correct thermodynamic equilibrium, is developed from activity measurements in the setup shown in Fig. 9.

The pressure drop over the catalyst is measured again with a charge from the test production, since the precise pellet size distribution and settling properties may differ from those of the lab or pilot scale production.

The new catalyst is now ready to be offered to the sulphuric acid industry.

11.4 Results

In the following, a selection of results from the development of VK69 are presented and discussed in order to illustrate the purpose and value of the individual steps of the scheme described above.

Numerous carriers were produced from different recipes and in different sizes and shapes in a 1 kg/min lab scale extruder and subsequently calcinated under different conditions in a furnace. The recipes included different types of diatomaceous earth, different types and amounts of binder and variation of the water content of the paste, which is a critical extrusion parameter. The shapes included among others rings, multiple-holed rings, finned rings, and trilobes, whereas normal cylindrical pellets were not made due to their well-known inferior activity to pressure drop ratio.

11.4.1 Activity Measurements

The carriers were impregnated and calcinated in the laboratory, and the activity was subsequently measured in the set-up shown in Fig. 9. The vanadium content was varied in the range 2-5 wt% and the molar ratios between alkali promoter and vanadium were varied in the ranges 0-4 for K/V, 0-2 for Na/V, and 0-3 for Cs/V. The sulphur content was about the same in all impregnations. The measured activities for 3 catalyst compositions A, B, and C impregnated on the same carrier and with the same vanadium content and molar ratios of K/V and Na/V are given in Table 1. The extrudates are made in the 9 mm Daisy form, which is the special 5-finned ring offered by Haldor Topsøe (Fig. 3). The observed pellet activity is reported as a pseudo-1st order rate constant calculated from

$$k_{1,obs} = -NHSV_{net} \cdot x_{eq} \cdot \ln(1 - x/x_{eq}) \quad (7)$$

where x_{eq} is the equilibrium conversion. According to the table, the Cs-containing catalysts are 61-117% more active than the standard product VK38 in the 12-mm Daisy form. The activity increases with Cs load, but seems to level off at high loadings. About 30% of the activity gain when compared to VK38 is due to the smaller size of the 9-mm Daisy particle.

Table 1. The effect of Cs content on the observed catalyst activity in a feed gas containing 0.7% SO₂ and 7% O₂.

Catalyst composition	A	B	C
Cs/V	low	medium	high
$NHSV_{net}$ (Nm ³ /h/m ³)	13089	14085	13228
Conversion, x	0.5972	0.6639	0.7023
$k_{1,obs}$ (Nm ³ /h/m ³)	11955	15446	16136
$k_{1,obs}/k_{I(VK38, 12\text{ mm})}$	1.61	2.08	2.17

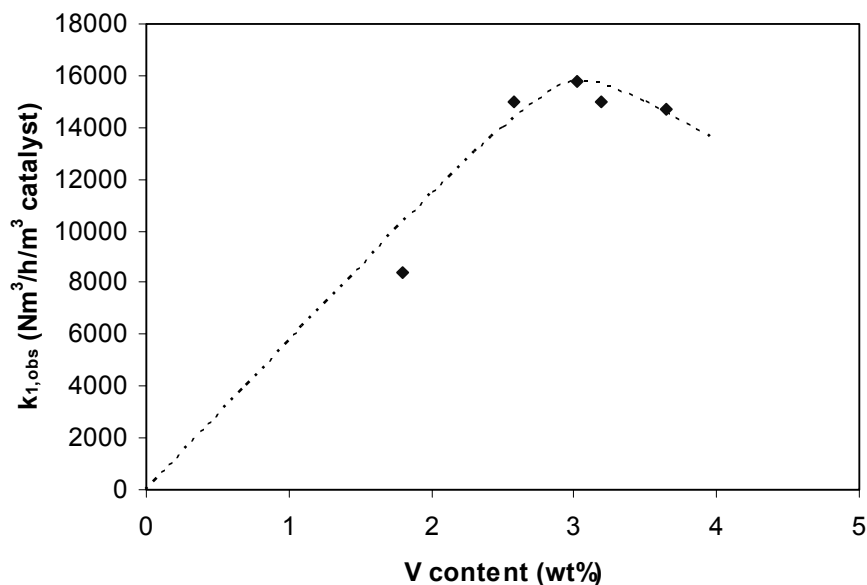


Figure 11. Observed activity of 1-2 mm catalyst particles in 0.7% SO₂ and 7% O₂ as a function of vanadium content for fixed molar ratios of K/V, Na/V, and Cs/V.

The effect of liquid loading was also studied by varying the V content at constant melt composition, i.e. constant ratios of K/V, Na/V, and Cs/V. An example of 1-2 mm large particles depicted in Fig. 11 shows an optimum vanadium content of about 3 wt% for this particular carrier and melt composition.

In order to optimize the size and shape of the catalyst particle, activities for pellets produced from the same material were measured for different sizes and shapes. Some of the results are reported in Table 2. The measured k_1 values show that 9-mm particles are 35% more active than 12-mm particles and that the activity of a 9-mm VK69 is 2.3 times better than a conventional VK38 in the form of a 12-mm Daisy.

The activity calculated from (7) comprises both film and pore diffusion resistance, but also the positive effect of increased temperature of the catalyst particle due to the exothermic reaction. From the observed reaction rates and mass- and heat transfer coefficients, it is found that the effect of external transport restrictions on the reaction rate is less than 5% in both laboratory and industrial plants. Thus, Table 2 shows that smaller catalyst particles are more active due to less diffusion restriction in the porous particle. For the dilute SO_2 gas, this effect can be analyzed by an approximate model assuming 1st order reversible and isothermal reaction. In this case, the surface effectiveness factor is calculated from

$$\eta_s = \frac{1}{\theta} \cdot \left(\frac{1}{\tanh(3\theta)} - \frac{1}{3\theta} \right) \quad (8)$$

with the Thiele modulus given by

$$\theta = \frac{d_{eq}}{6} \sqrt{\frac{k_1}{D_{eff} \cdot x_{eq}}} \quad (9)$$

The activity in terms of 1st order rate constant $k_{1,calc}$ was calculated in Table 2 from (8) and (9) with effective diffusivity $D_{eff}=5.3 \cdot 10^{-6} \text{ m}^2/\text{s}$ and intrinsic rate constant $k_1=33000 \text{ Nm}^3/\text{h/m}^3 = 23 \text{ s}^{-1}$ fitted to the measurements. This simple and useful method models the measured influence of particle size satisfactorily for a first optimization of particle size and shape. The 35% higher activity measured for the 9-mm Daisy compared to the 12-mm Daisy, however, exceeds the 25% expected from (8), and this illustrates the importance of measuring the activity of the actual shape.

The 1-2 mm particles were prepared by crushing the 9 mm pellets and separating the 1-2 mm particle size fraction by sieving. In this way, the small particles have the same intrinsic properties as the 9 mm pellet. However, consistent results are not always obtained because a larger fraction of the

catalyst volume comes from the interior of a pellet, which is different from the surface layer due to the processes taking place during extrusion, drying and calcination. In order to obtain truly intrinsic activity ($\eta \approx 1$), even smaller particles may be tested, but for very small particles ($< \text{about } 0.2 \text{ mm}$) channelling and gas bypass may be a problem.

For a more detailed analysis of measured transport restrictions and reaction kinetics, a more complex reactor simulation tool developed at Haldor Topsøe was used. The model used for sulphuric acid catalyst assumes plug flow and integrates differential mass and heat balances through the reactor length [16]. The bulk effectiveness factor for the catalyst pellets is determined by solution of differential equations for catalytic reaction coupled with mass and heat transport through the porous catalyst pellet and with a film model for external transport restrictions. The model was used both for optimization of particle size and development of intrinsic rate expressions. Even more complex models including radial profiles or dynamic terms may also be used when appropriate.

Table 2. The effect of catalyst pellet size on the observed activity of VK69 in feed gas containing 0.7% SO₂ and 7% O₂.

Pellet size	10 mm	12 mm	9 mm	1-2 mm
Shape	Ring	Daisy	Daisy	crushed
$d_{eq} \text{ (mm)}$	7.07	6.25	4.68	1.3
$NHSV_{net} \text{ (Nm}^3\text{/h/m}^3\text{)}$	13053	12771	16393	20080
Conversion, x	0.5813	0.6298	0.6485	0.775
$k_{l,obs} \text{ (Nm}^3\text{/h/m}^3\text{)}$	11412	12755	17233	30233
$k_{l,obs}/k_{l(VK38, 12 \text{ mm})}$	1.54	1.72	2.32	4.07
θ	2.41	2.13	1.60	0.44
η_s	0.36	0.40	0.50	0.90
$k_{l,calc} = \eta_s \cdot k_l \text{ (Nm}^3\text{/h/m}^3\text{)}$	11801	13068	16368	29669

11.4.2 Mechanical Strength Results

The mechanical strength of the carriers produced in lab scale (Fig. 8) was quantified in terms of attrition loss, side crush strength and drop test strength. An example of the results is given in Table 3 for carriers D and E prepared with

different amounts of binder in the recipe. Pellets D made with a high amount of binder became more dense with a resulting higher side crush strength and a lower fractional break-up in the drop tests. However, the lower effective diffusivity for the dense pellets also resulted in about 40% lower activity.

In order to compare the strength of different catalyst shapes produced from the same recipe, drop tests were carried out for 10-mm rings and 12-mm Daisy shapes in Table 4. It is evident from the results that Daisy pellets are much more resistant to break-up in the drop test than the rings. The difference may be explained by assuming that the ribbons on the Daisy shape protect the cylinder against breakage. Although the rings are heavier than the Daisy pellets, this seems to be of less importance considering that the falling times are almost the same. It was also observed, that the material in the ribbons was harder than that deeper inside the cylinder and that the ribbons were not even damaged when acting as shock absorbers.

Table 3. Strength test results for two carriers produced as 12 mm Daisy from similar recipes except for the amount of binder.

Carrier recipe	D	E
Amount of binder	high	low
Pellet density (g/ml)	0.74	0.62
Side crush strength (avg. for 10 pellets, N/cm pellet length)	67 ± 8	24 ± 5
<u>Drop test (20 pellets, 2 tests)</u>		
- No of broken pellets (test 1/test 2)	0 / 1	2 / 2
- weight loss (test 1/test 2)	4.4 / 5.0	7.5 / 5.5

Table 4. Drop test results for unused catalysts produced from the same recipes but in different shapes.

Sample	Shape	Fall time (s)	Broken/total	%broken
1	10-mm ring	2.03	12 / 20	60
2	10-mm ring	1.99	21 / 31	68
3	12-mm Daisy	2.15	3 / 20	15
4	12-mm Daisy	2.15	10 / 45	22

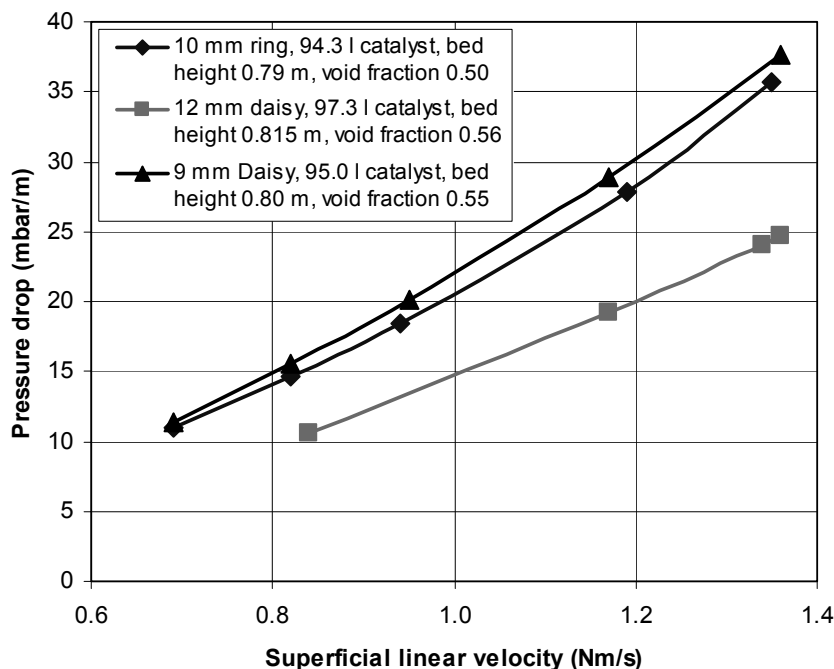


Figure 12. Measured pressure drop for cylindrical rings and Daisy-shapes

11.4.3 Pressure Drop Measurements

The pressure drop for different catalyst shapes (10 mm cylindrical rings, 12 mm Daisy particles and 9 mm Daisy particles) was measured in the set-up shown in Fig. 10 as a function of superficial linear velocity (Fig. 12). The pressure drop per bed height is about 30% lower for a 12 mm Daisy compared to a 10-mm ring primarily due to the lower void of the bed with 10-mm rings (0.50 compared to 0.56). The 9-mm Daisy yields a 5% higher pressure drop than the rings because the higher void is counteracted by the smaller equivalent diameter. It should be noted that the pressure drop of the 12-mm Daisy is only 66% of the 9-mm pressure drop, i.e. lower than the $9/12 = 75\%$ expected from the Ergun equation (5) for equal shapes. The deviation is attributed to the difference in void fraction, which arises from the difference in distribution of particle lengths and in the actual geometry of the pellet ends. This observation illustrates that the theoretical correlations only can be used as a steering tool or an interpolation in the development and that the measured pressure drops are necessary for proper comparison.

The screening of carriers, catalyst composition, particle sizes and shapes showed indeed, that a much more active catalyst could be made with Cs as a secondary promoter for the beds downstream the intermediate absorption tower. The best candidates were selected, and some m³ of each recipe were produced as 9-mm and 12-mm Daisy extrudates in a successful commercial-scale test production. The activities were as expected from the previous development work, and a 30 day activity test also confirmed it to be stable during this period.

11.4.4 Industrial Stability Tests

A canister with several VK69 formulations was also installed in the 4th bed of an industrial 3+1 converter in a 2500 t/day sulphuric acid plant treating off-gas from a copper smelter. After two years of operation at 415°C inlet temperature and 0.7-1% SO₂, the canister was retrieved and the catalysts were analysed as exemplified in Table 5 for one of the candidates with concentrations normalised to the unused catalyst. The results were very satisfactory with unchanged side crush strength and very low deactivation. The small drop observed for the vanadium concentration is primarily due to the 5% weight gain of the catalyst caused by SO₃ uptake according to reaction (2), and the small changes in alkali-metal-to-vanadium ratios were considered to be within the uncertainty of the test. Additional valuable analyses, e.g. elemental composition by ICP and analysis of solid phases by X-ray diffraction, were also carried out but not given in the table. Furthermore, the pore size distribution of the carrier was measured by leaching the melt from the catalyst prior to Hg porosimetry analysis. These analyses provide information on the operating history, potential catalyst poisons, and carrier stability.

Table 5. Results for a VK69 formulation before and after two years of industrial operation.

	Unused catalyst	0-11 cm from top (after 2 years)	23-33 cm from top (after 2 years)
V (relative wt%)	1	0.89	0.97
K/V (relative)	1	0.93	0.93
Na/V (relative)	1	1.02	1.02
Cs/V (relative)	1	1.03	1.04
S/(K+Na+Cs) (molar ratio)	0.77	0.94	0.97
Activity (relative)	1	0.95	0.96
Side crush strength, kp/cm	7	7	8

11.5 Evaluation of Results

The evaluation of carriers and catalyst compositions showed that significantly higher SO₂ oxidation activity could be achieved with Cs as a promoter under the operating conditions downstream the intermediate absorption tower as demonstrated by the results in Table 1, where the activity compared to the standard product is increased by more than a factor 2. This was clearly sufficient for the introduction of VK69 to the market as a new sulphuric acid catalyst. The activity results for different melt compositions were used to optimise the vanadium content and the molar ratios of K/V, Na/V, and Cs/V. However, the choice of Cs/V was not only a question of maximum activity, because of the significant influence of the Cs content on the raw material costs (the price of caesium is 50-100 times the price of potassium on a molar basis). Here, the economic benefits obtained by the sulphuric acid producer by the marginal activity improvement at high Cs content also had to be taken into account.

The laboratory tests as well as the industrial canister tests demonstrated excellent long-term stability of both activity and mechanical strength for the VK69 formulations selected for the tests. For the optimisation of pellet size and shape, the results for activity, fractional void, and pressure drop must be compared as shown in Table 6 for 10 mm rings, 12 mm Daisy-shape, and 9 mm Daisy-shape.

The basis of the comparison is the catalyst bed volume (reactor volume) required to achieve a given SO₂ conversion. The volume is inversely proportional to the bulk activity, which in turn is given by equation (4). For this bed volume, the pressure drop and the net catalyst volume accounting for a fraction of $(1-\epsilon)$ of the bed volume are compared for the three pellet types. The 12-mm Daisy-shape requires about the same bed volume as the 10-mm ring but the pressure drop can be reduced by 29%. The 9-mm Daisy-shape requires 23% less bed volume than the 10-mm ring giving 19% less pressure drop. The required catalyst volume is important in itself for plants where the SO₂ conversion is limited by the volume of the 4th reactor bed, i.e. the SO₂ emission can only be lowered by installing a more active catalyst and not by adding more catalyst. Above some minimum pressure drop required for even gas flow distribution, a lower pressure drop is always desired but in particular for plants with air blowers operating at their maximum capacity. The significance of the required amount of catalyst material (net pellet volume) is the raw material

consumption - in the case of VK69 especially the more expensive Cs. Because the main market for the new catalyst was expected to be lowering of SO₂ emissions and boosting of production capacity, the more active 9-mm Daisy shape was chosen as the primary product design. For plants with pressure drop limitations, the bed can be loaded with the 12-mm Daisy. The 10-mm ring was ruled out due to the superior activity to pressure drop ratio of the Daisy-shapes.

The comparison in Table 6 illustrates the method but it is only an approximate comparison. In order to account for differences in apparent activation energy between the catalysts, a full reactor model integrating the reaction rate over the full temperature range of the bed is necessary for calculation of the exact catalyst volume.

Table 6. Optimisation of pellet size and shape. The catalyst volume required to obtain a specified SO₂ conversion is the basis for comparison of performance.

Pellet size	10 mm	12 mm	9 mm
Shape	Ring	Daisy	Daisy
Measured pellet activity (relative)	1	1.12	1.45
Measured pressure drop per m bed (relative)	1	0.70	1.05
Measured bed void fraction	0.50	0.56	0.55
Bulk activity (relative)	1	0.99	1.31
Required cat. bed volume (relative)	1	1.01	0.77
Bed pressure drop (relative)	1	0.71	0.81
Net pellet volume (relative)	1	0.89	0.69

11.6 Conclusions

The development of catalysts includes both practical and theoretical knowledge and experience with a palette of disciplines known to the chemical engineer, e.g. inorganic chemistry, physical and chemical characterisation techniques, nano-scale materials, unit operations, chemical reaction engineering, catalyst kinetics, and transport phenomena.

The design of the new sulphuric acid catalyst VK69 by Haldor Topsøe was based on experience from existing products, market knowledge, and an innovative attitude especially in the first stages of development. The

development work did not strictly follow the systematic approach outlined in Fig. 7, because a lot of experience from previous tests and development projects including e.g. identification of critical parameters made it possible to cut corners and bypass steps for some of the VK69 candidates. A systematic and full parametric investigation in each step may strengthen the competencies involved, but from an economic point of view, a more direct route to the goal is preferred because this will reduce the development cost and the time to product launch.

VK69 was introduced in the market by Haldor Topsøe in 1996 and has subsequently been installed in the final passes of more than 50 double absorption sulphuric acid plants worldwide. Due to the superior activity of VK69, the SO₂ emissions can typically be reduced by a factor 2 and new double absorption plants can be designed with less than 40 ppm SO₂ emission [17]. The industrial experience also confirms low deactivation rate and low screening loss as expected from the results obtained in the development phase.

Nomenclature

A_p	Pellet external surface area [m ²]
d_{eq}	Equivalent pellet diameter $=6 \cdot V_p/A_p$ [m]
D_{eff}	Effective diffusion coefficient in the catalyst
k_1	First-order rate constant [Nm ³ gas/h/m ³ catalyst pellet]
Nm ³	Normal m ³ gas at 0°C and 1 atm.
Nm	Nm ³ /m ²
$NHSV$	Normal hourly space velocity [Nm ³ /h/m ³ catalyst bed]
$NHSV_{net}$	Normal hourly space velocity [Nm ³ /h/m ³ catalyst pellet]
P	Total pressure [Pa]
p_i	Partial pressure of component i [Pa]
r_{SO_2}	Rate of SO ₂ production [mole/s/m ³ catalyst pellet]
T	Temperature [°C]
V_p	Pellet volume [m ³]
x	Conversion of SO ₂ [-]
x_{eq}	Equilibrium conversion of SO ₂ [-]
ϵ	Bed void fraction
θ	Thiele modulus
η_b, η_s	Bulk or surface effectiveness factor

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